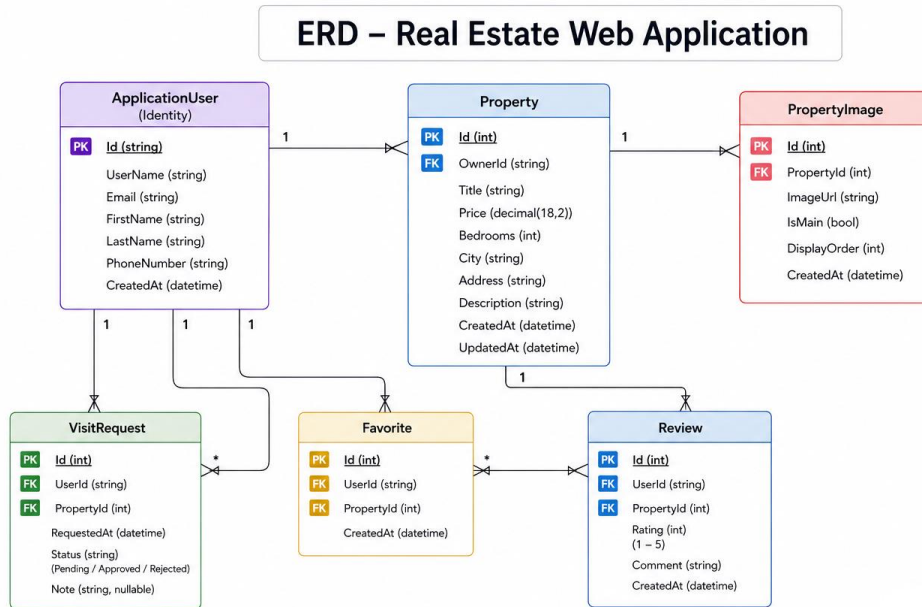


Database Design & Data Modeling

ER Diagram (Entity-Relationship Diagram) – A well-defined ERD showcasing database structure and relationships.

1. Erd Diagram



Logical & Physical Schema – Tables, attributes, keys, and normalization considerations.

1. Logical Schema (Database Design)

This section defines the database structure in a conceptual way, including tables, attributes, and relationships.

ApplicationUsers

- Id (PK)
- UserName
- Email
- FirstName
- LastName

- PhoneNumber
- CreatedAt

Properties

- PropertyId (PK)
- Title
- Description
- Price
- City
- Address
- Bedrooms
- CreatedAt
- UpdatedAt
- OwnerId (FK → ApplicationUsers.Id)

PropertyImages

- ImageId (PK)
- PropertyId (FK → Properties.PropertyId)
- ImageUrl
- IsMain (bool)
- DisplayOrder
- CreatedAt

VisitRequests

- RequestId (PK)
- UserId (FK → ApplicationUsers.Id)
- PropertyId (FK → Properties.PropertyId)
- RequestedAt
- Status (Pending / Approved / Rejected)
- Note (nullable)

Favorites

- FavoriteId (PK)
- UserId (FK → ApplicationUsers.Id)
- PropertyId (FK → Properties.PropertyId)
- CreatedAt

Reviews

- ReviewId (PK)
- UserId (FK → ApplicationUsers)
- PropertyId (FK → Properties)
- Rating
- Comment
- CreatedAt

2. Physical Schema (SQL Implementation)

```
CREATE TABLE ApplicationUsers (  
    Id NVARCHAR(450) PRIMARY KEY,  
    UserName NVARCHAR(100),  
    Email NVARCHAR(256) UNIQUE NOT NULL,  
    FirstName NVARCHAR(100),  
    LastName NVARCHAR(100),  
    PhoneNumber NVARCHAR(20),  
    CreatedAt DATETIME DEFAULT GETDATE()  
);
```

```
CREATE TABLE Properties (  
    PropertyId INT PRIMARY KEY IDENTITY,  
    Title NVARCHAR(150) NOT NULL,  
    Description NVARCHAR(MAX),  
    Price DECIMAL(18,2) NOT NULL,  
    City NVARCHAR(100),  
    Address NVARCHAR(200),  
    Bedrooms INT,  
    CreatedAt DATETIME DEFAULT GETDATE(),  
    UpdatedAt DATETIME,  
    OwnerId NVARCHAR(450),  
    FOREIGN KEY (OwnerId) REFERENCES ApplicationUsers(Id)  
);
```

```
CREATE TABLE PropertyImages (  
    ImageId INT PRIMARY KEY IDENTITY,  
    PropertyId INT,  
    ImageUrl NVARCHAR(500),  
    IsMain BIT DEFAULT 0,  
    DisplayOrder INT,
```

```
    CreatedAt DATETIME DEFAULT GETDATE(),  
  
    FOREIGN KEY (PropertyId) REFERENCES Properties(PropertyId)  
  
);
```

```
CREATE TABLE VisitRequests (  
  
    RequestId INT PRIMARY KEY IDENTITY,  
  
    UserId NVARCHAR(450),  
  
    PropertyId INT,  
  
    RequestedAt DATETIME DEFAULT GETDATE(),  
  
    Status NVARCHAR(20) DEFAULT 'Pending',  
  
    Note NVARCHAR(500),  
  
    FOREIGN KEY (UserId) REFERENCES ApplicationUsers(Id),  
  
    FOREIGN KEY (PropertyId) REFERENCES Properties(PropertyId)  
  
);
```

```
CREATE TABLE Favorites (  
  
    FavoriteId INT PRIMARY KEY IDENTITY,  
  
    UserId NVARCHAR(450),  
  
    PropertyId INT,  
  
    CreatedAt DATETIME DEFAULT GETDATE(),  
  
    FOREIGN KEY (UserId) REFERENCES ApplicationUsers(Id),
```

```
FOREIGN KEY (PropertyId) REFERENCES Properties(PropertyId)  
);
```

```
CREATE TABLE Reviews (  
    ReviewId INT PRIMARY KEY IDENTITY,  
    UserId NVARCHAR(450),  
    PropertyId INT,  
    Rating INT CHECK (Rating BETWEEN 1 AND 5),  
    Comment NVARCHAR(500),  
    CreatedAt DATETIME DEFAULT GETDATE(),  
    FOREIGN KEY (UserId) REFERENCES ApplicationUsers(Id),  
    FOREIGN KEY (PropertyId) REFERENCES Properties(PropertyId)  
);
```

3. Normalization Considerations

The database is normalized up to Third Normal Form (3NF) to ensure data consistency, reduce redundancy, and improve efficiency.

1NF: Ensures atomic values and no repeating groups.

2NF: Ensures no partial dependency on composite keys.

3NF: Ensures no transitive dependencies between non-key attributes.